Name:

Quiz 2 Solutions

Instructions. The only tools you are permitted to use are pencils, pens, erasers, a sheet of handwritten notes, and your mind. No electronic devices! You have 1 hour. Be sure to sign the Honor Code statement on the last page. Good luck!

Problem 1 (True-false, 8 points)**.** You will get 1 point for a correct answer and 0 points for blank and incorrect answers.[1] No explanations are required.

(1) The system $x' = xy$ and $y' = x + y$ is linear. T **F**

(2) The solutions to the system $\vec{x}\,' = \begin{bmatrix} 0 & 2t \\ 0 & 2t \end{bmatrix} \vec{x}$ form a 2-dimensional vector space. **T** F

(3) The vector-valued functions $\begin{bmatrix} e^{3t} \\ e^{t} \end{bmatrix}, \begin{bmatrix} e^{t} \\ e^{2t} \end{bmatrix}$ are linearly independent. **T** F

(4) If A is a 2×2 matrix whose entries are real numbers and $\begin{bmatrix} 3+2i \\ 5 \end{bmatrix}$ is an eigenvector of A with eigenvalue $-1+2i$, then $\begin{bmatrix} 3-2i \\ 5 \end{bmatrix}$ is an eigenvector of A with eigenvalue $-1-2i$. **T** F

(5) Suppose $x''' = x + x' + \sin(t)$. If $\vec{x} = \begin{bmatrix} x \\ x' \\ x'' \end{bmatrix}$, then $\vec{x}\,' = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \vec{x} + \begin{bmatrix} 0 \\ 0 \\ \sin(t) \end{bmatrix}$. **T** F

(6) The solution space of the ODE $x''' - 3x'' + 4x' - 12x = 0$ is spanned by the functions $e^{3t}, \sin(2t), \cos(2t)$. **T** F

(7) If $-1+i$ is an eigenvalue of a 2×2 matrix A and $\begin{bmatrix} 4+i \\ 5 \end{bmatrix}$ is a corresponding eigenvector, the function $\vec{x}(t) = e^{-t} \begin{bmatrix} 4\cos(t) \\ 5\cos(t) \end{bmatrix}$ must be a solution of the system $\vec{x}\,' = A\vec{x}$. T **F**

(8) The phase portrait of the system $\vec{x}\,' = \begin{bmatrix} 0 & 1 \\ -1 & a \end{bmatrix} \vec{x}$ is a spiral source if and only if[2] $-2 < a < 2$. T **F**

[1] During quiz revisions, you'll have the chance to meet with me one-on-one and convince me that you understand and can fully explain *up to 2 of the true-false questions that you left blank* to get full credit on those questions.

[2] In case you haven't seen the "if and only if" terminology before, the assertion being made is two-fold: (1) that the phase portrait is a spiral source if a is strictly between -2 and 2, and (2) that the phase portrait is *not* a spiral source if a is *not* strictly between -2 and 2.

Problem 2 (2 points)**.** Calculate the general solution of the first order linear system

$$\frac{d\vec{x}}{dt} = \begin{bmatrix} 2 & -1 \\ -3 & 0 \end{bmatrix} \vec{x}.$$

Solution. The characteristic polynomial is

$$-\lambda(2-\lambda) - 3 = \lambda^2 - 2\lambda - 3 = (\lambda - 3)(\lambda + 1),$$

so the eigenvalues are $\lambda = -1, 3$. An eigenvector corresponding to $\lambda_1 = -1$ is a nonzero vector in the null space of

$$A + I = \begin{bmatrix} 3 & -1 \\ -3 & 1 \end{bmatrix} \to \begin{bmatrix} 3 & -1 \\ 0 & 0 \end{bmatrix}$$

Thus $\vec{v}_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ spans the eigenspace. Similarly, the eigenspace corresponding to $\lambda_2 = 3$ is the null space of

$$A - 3I = \begin{bmatrix} -1 & -1 \\ -3 & -3 \end{bmatrix} \to \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}.$$

Thus $\vec{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ spans the eigenspace. Thus the general solution to the system is

$$\vec{x}(t) = C_1 e^{-t} \begin{bmatrix} 1 \\ 3 \end{bmatrix} + C_2 e^{3t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Problem 3 (2 points)**.** Calculate the general solution of the ODE $x'' + 4x' + 20x = 0$.

Solution. We guess $x(t) = e^{rt}$ and find that $r^2 + 4r + 20 = 0$, which means that

$$r = \frac{-4 \pm \sqrt{16 - 80}}{2} = -2 \pm 4i.$$

Thus

$$e^{(-2+4i)t} = e^{-2t}e^{4it} = e^{-2t}(\cos(4t) + i\sin(4t))$$

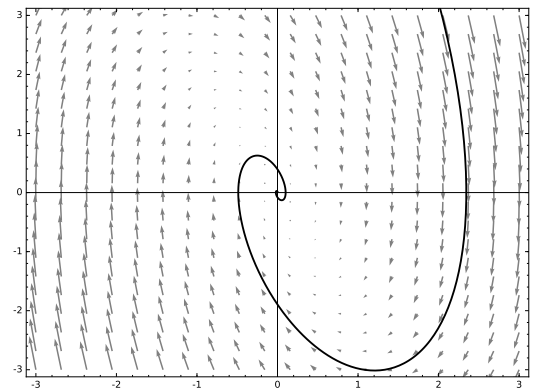
is a complex-valued solution. Taking real and imaginary parts, we find that $e^{-2t}\cos(4t)$ and $e^{-2t}\sin(4t)$ are both solutions. They are linearly independent, so they must span the solution space. In other words, any real solution must be of the form

$$x(t) = C_1 e^{-2t}\cos(4t) + C_2 e^{-2t}\sin(4t)$$

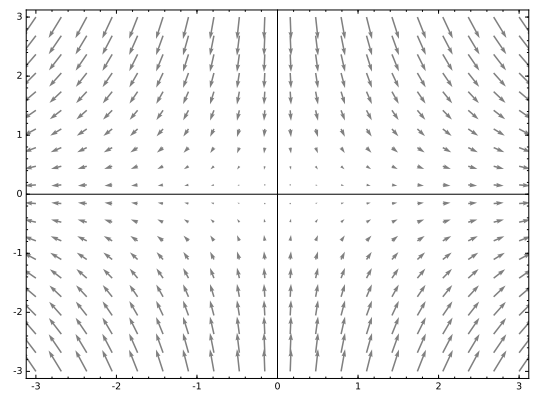
for real constants C_1 and C_2.

Problem 4 (3 points)**.** On the left are descriptions of several 2×2 matrices. On the right are several phase portraits. Match each matrix A with the phase portrait of the system $\vec{x}\,' = A\vec{x}$.

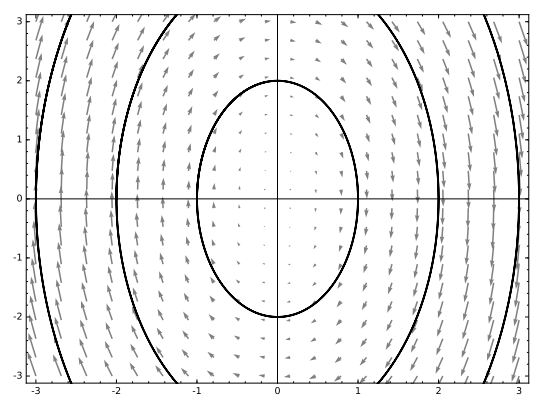
(a) $A = \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix}$



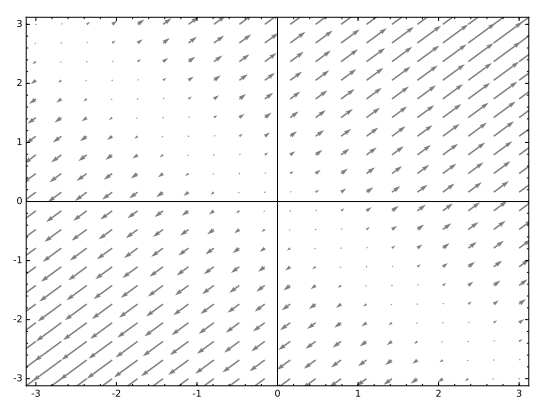
(b) A is a matrix with eigenvalues $-1 \pm 2i$



(c) A has eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 0$, with corresponding eigenvectors $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$, respectively.



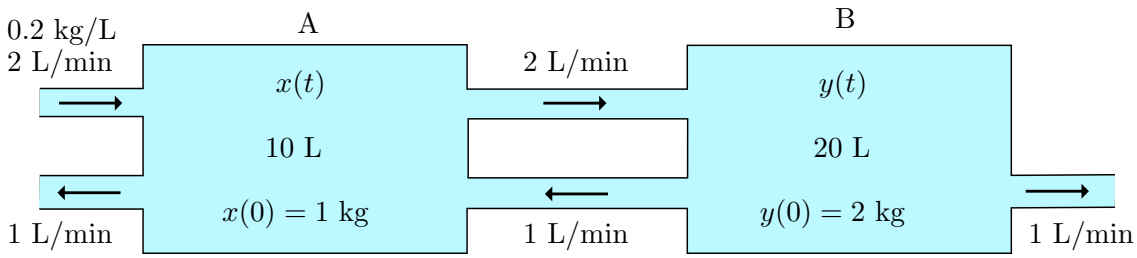
(d) $A = \begin{bmatrix} 0 & 1 \\ -4 & 0 \end{bmatrix}$



Solution. In the order of the matrices, the phase portraits are (b), (a), (d), (c).

Problem 5 (3 points)**.** Nope, it's not salt water tanks. It's sugar water tanks :P

Suppose there are two interconnected tanks A and B with sugar water flowing between them as depicted in the diagram below. Suppose further that the input coming into tank A from outside has a constant sugar concentration of 0.2 kg/L. Let $x(t)$ and $y(t)$ denote the quantities of sugar that are dissolved in tanks A and B at time t, respectively. Assume perfect mixing.



(a) Find a matrix A and a vector-valued function $\vec{f}$ such that

$$\frac{d}{dt}\begin{bmatrix} x \\ y \end{bmatrix} = A\begin{bmatrix} x \\ y \end{bmatrix} + \vec{f}.$$

You do *not* need to solve the system.

Solution. We have

$$\frac{dx}{dt} = -\frac{3x}{10} + \frac{y}{20} + 0.4 \quad \text{and} \quad \frac{dy}{dt} = \frac{2x}{10} - \frac{2y}{20}.$$

Thus

$$\frac{d}{dt}\begin{bmatrix} x \\ y \end{bmatrix} = \underbrace{\begin{bmatrix} -3/10 & 1/20 \\ 2/10 & -2/20 \end{bmatrix}}_{A}\begin{bmatrix} x \\ y \end{bmatrix} + \underbrace{\begin{bmatrix} 0.4 \\ 0 \end{bmatrix}}_{\vec{f}}$$

(b) Use physical intuition to guess the value of $\lim_{t\to\infty}\begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$.

Solution. One expects that, over time, both tanks will end up with a sugar concentration of 0.2 kg/L. This means that tank A will have 2 kg of sugar and tank B will have 4 kg of sugar, so

$$\lim_{t\to\infty}\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}.$$

Problem 6 (Extra credit! 2 points, no partial credit)**.** Construct a matrix A such that the general solution to $\vec{x}\,' = A\vec{x}$ is given by

$$\vec{x}(t) = C_1 e^{-t} \begin{bmatrix} -\sin(2t) \\ 2\cos(2t) \end{bmatrix} + C_2 e^{-t} \begin{bmatrix} \cos(2t) \\ 2\sin(2t) \end{bmatrix}$$

Explain your process.

Solution. Hint: diagonalization! A complete solution is omitted so you can have fun thinking about it :) Feel free to come talk to me about it if you're curious and stuck!

Honor code. If you have neither given nor received any unauthorized aid on this quiz, please write either "HCU" or "Honor Code Upheld" below, and sign your name next to it.

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